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22SW - Initial Seminar – Comments & Suggestions

All students of 22SW are obligated to consider and adhere to the feedback concerning their FYP initial seminars, and to implement all such recommendations for their subsequent progressive seminars.

22FYDP01 – CareerCompass

Roll Nos.: 22SW34, 22SW20, 22SW59
Supervisor: Prof. Dr. Pardeep Kumar
Comments / Suggestions:

- Narrow the project scope to predictions that are realistically achievable with the available dataset.
- Clearly define the system's prediction output (e.g., job role recommendations, skill-gap identification).
- Prepare a detailed plan for dataset collection, cleaning, preprocessing, and labeling.
- Specify evaluation metrics such as accuracy, precision, recall, and F1-score.
- Include transparency and fairness components (e.g., feature importance, bias assessment).
- Avoid broad/global claims unless supported by genuine large-scale/global datasets.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP02 – Mazdoor App

Roll Nos.: 22SW11, 22SW25, 22SW36
Supervisor: Prof. Dr. Pardeep Kumar
Comments / Suggestions:

- Refine the problem statement by addressing issues like laborer verification, trust, safety, and communication gaps.
- Clearly define the scope and functionality of each portal:
    - **Laborer App:** Registration, profile, job search, acceptance, payments, ratings.

  - **Client App:** Job posting, worker matching, hiring, payment, feedback.
  - **Admin Dashboard:** User monitoring, logs, dispute handling, analytics.
- Include a plan for laborer identity and skill verification (ID check, certificates, ratings).
- Clarify the secure payment method (gateway integration, wallet, or cash-on-completion).
- Ensure compliance with local labor laws and policies.
- Add measurable evaluation criteria such as usability testing, matching accuracy, and performance benchmarks.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP03 – RideMate: Smart Ride Sharing App

Roll Nos.: 22SW24, 22SW39, 22SW58
Supervisor: Dr. Imtiaz Halepoto
Comments / Suggestions:

- Specify the geographical scope (single campus, city-wide, or multiple areas).
- Integrate safety mechanisms: verified drivers, vehicle checks, live GPS tracking, SOS alerts.
- Strengthen user verification through phone, email, or student ID authentication.
- Define core features: ride creation, finding, booking, cancellation, recurring rides.
- Add filters and sorting options (cost, time, rating).
- Implement a two-way rating and feedback system.
- Ensure secure online payments and digital receipt generation.
- Integrate ride history with payments and feedback.
- Ensure system stability during peak hours, holidays, or local city disturbances.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP04 – QUESTPAY: Smart Fee Management App

Roll Nos.: 22SW32, 22SW40, 22SW57
Supervisor: Dr. Imtiaz Halepoto
Comments / Suggestions:

- Explain how the system integrates with the finance department's workflow while maintaining accuracy and transparency.
- Clarify technical processes such as QR verification, receipt validation, and controlled admin updates.
- Define secure authentication, audit trails, admin authorization levels, and handling of fee variations (fines, corrections, concessions).
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP05 – Digital Marketplace with AI-Powered Virtual Try-On (VTO)

Roll Nos.: 22SW31, 22SW37, 22SW06
Supervisor: Dr. Fayaz Ahmed Memon
Comments / Suggestions:

- Start with the most feasible VTO item (e.g., glasses or simple shirts) before moving to complex apparel.
- Provide a technical fallback plan in case the Kling AI API becomes too costly or slow.
- Define the product data sellers must upload (e.g., 3D models, multi-angle images).
- Define how VTO "accuracy" will be measured to show improvement in customer confidence or reduction in returns.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP06 – JOBVERSE Android App

Roll Nos.: 22SW35, 22SW04, 22SW05
Supervisor: Dr. Fayaz Ahmed Memon
Comments / Suggestions:

- Clarify how job listings and expert searches will be localized to the user's immediate vicinity.

- Explain how experts' skills, experience, and quality will be verified.
- Identify measures to prevent spam, misuse, or negative interactions.
- Specify design decisions that will support future features like ratings, payments, and advanced matching.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP07 – Accident Detection & Alert System Using Smartphone Sensors

Roll Nos.: 22SW47, 22SW29, 22SW18
Supervisor: Dr. Rafia Naz
Comments / Suggestions:

- Address the risk of false triggers (e.g., device drops, sudden braking). Define a multi-sensor confirmation strategy (accelerometer, gyroscope, GPS).
- Clarify how accurate GPS location will be captured even under weak signal conditions.
- Define the complete alert chain:
    1. Reliable accident detection
    2. Automatic confirmation timer
    3. GPS capture
    4. SMS alert to a preset contact
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP08 – Plant Disease Monitoring Using CNN

Roll Nos.: 22SW52, 22SW51, 22SW14
Supervisor: Dr. Sajida
Comments / Suggestions:

- Identify the dataset source — public or locally collected wheat disease images.
- Limit to a manageable number of disease classes (e.g., rust, smut, healthy).
- Clarify whether inference will be offline (on-device) or online (server-based).
- Explain image processing speed and expected response time.

- Define validation procedure for unseen test images.
- Clarify whether multiple diseases can be detected in a single image.
- Improve the literature review and comparative summary table and Use proper IEEE reference style.

22FYDP09 – Iris Detection System for Medical Disease Diagnosis

Roll Nos.: 22SW50, 22SW03, 22SW38
Supervisor: Dr. Sajida
Comments / Suggestions:

- Specify one focused disease category for the project.
- Define how high-quality, clinically labeled iris datasets will be obtained.
- Clearly state the required hardware (macro lens, resolution specs).
- Explain how lighting and reflection issues will be handled.
- Provide strategies for reducing false negatives.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP10 – Latif Ji Boli: A Sindhi Programming Language

Roll Nos.: 22SW30, 22SW12, 22SW43
Supervisor: Dr. Ali Raza Bhangwar
Comments / Suggestions:

- Finalize Sindhi keywords and their English/Java equivalents early to avoid re-writing lexical and parsing components.
- Identify pipeline components that require specialized tooling (ANTLR, Flex/Bison).
- Explain how the AI component will be trained to debug Sindhi-language programming errors.
- Prepare a complete Sindhi-language documentation base (keywords, error messages, examples) for the RAG system.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP11 – Q-Oribot: Smart AI Assistant for QUEST Students

Roll Nos.: 22SW07, 22SW46, 22SW49
Supervisor: Engr. Muhammad Aamir Bhuto
Comments / Suggestions:

- Define validation methods for verifying AI-generated responses to institutional queries.
- Use a mobile-first design for the React frontend to ensure accessibility.
- Design a scalable knowledge base structure (e.g., JSON) supporting bilingual content.
- Define procedures to ensure accuracy of the knowledge base and assign responsibility for future updates.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP12 – QUTA: Blockchain-Based Voting System

Roll Nos.: 22SW33, 22SW44, 22SW17
Supervisor: Engr. Muhammad Aamir Bhuto
Comments / Suggestions:

- Address Ethereum gas fees and plan cost/time optimization for conducting elections.
- Identify the most critical security features for the centralized dashboard used by the Election Commissioner.
- Ensure that the imported teacher/voter list is complete, validated, and protected from tampering.
- Verify the smart contract for correctness, ensuring prevention of double voting and accurate election closure.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP13 – Guesthouse Booking & Room Services

Roll Nos.: 22SW21, 22SW22, 22SW41
Supervisor: Dr. Sajida
Comments / Suggestions:

- Identify the simplest training dataset and explain how recommendations will create real operational value.

- Clarify the practical need and benefit of AI-based room recommendation versus simple availability listings.
- Provide a low-cost hosting and maintenance plan for long-term usability.
- Ensure the system handles cancellations, secure payments, audit logs, admin authorization, and meaningful data for AI training.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP14 – Data-Driven Software Solution for Customer Retention & Product Insights

Roll Nos.: 22SW02, 22SW13, 22SW48
Supervisor: Mr. Zohaib Dahri
Comments / Suggestions:

- Validate relevance of Kaggle datasets by comparing them with local e-commerce trends.
- If needed, synthesize or weight local factors to make predictions regionally relevant.
- Final dashboard must provide actionable insights (e.g., "Customer X has 85% churn risk — send a 10% discount coupon").
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP15 – Hybrid AI for Recruitment

Roll Nos.: 22SW01, 22SW27, 22SW56
Supervisor: Mr. Zohaib Dahri
Comments / Suggestions:

- Standardize all resume formats and handle missing or inconsistent data fields.
- Apply noise reduction, normalization, and data augmentation for audio/video inputs.
- Define fairness metrics and introduce bias reduction strategies.
- Create a unified scoring system combining hard and soft skill evaluations.
- Justify the selected NLP and CNN models.
- Utilize pre-trained models to enhance performance and save development time.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP16 – Smart AI-Based Academic Management System (SAMS)

Roll Nos.: 22SW55, 22SW53, 22SW15
Supervisor: Mr. Zohaib Dahri
Comments / Suggestions:

- The "Smart" in SAMS primarily relies on face recognition for attendance. While useful, this is a transactional automation, not true "smart" intelligence.
- Integrate modules for predictive analytics. For instance:
- **Early Warning System:** Analyze attendance patterns, assignment submissions, and quiz scores to predict students at risk of failing. The system could alert advisors or provide personalized intervention suggestions.
- **Course Recommendation:** Based on student performance in prerequisite courses, interests, and career goals (data points collected within SAMS), recommend advanced courses or relevant electives.
- **Resource Recommendation:** Link underperforming students to specific learning resources (e.g., tutorials, extra practice problems) identified by the system based on their weaknesses.
- Improve the literature review and comparative summary table and use proper IEEE reference style.
- **Face Recognition Accuracy:** Report metrics like precision, recall, F1-score, and false acceptance/rejection rates. Test under various conditions (lighting, angles, occlusions).
- **Qualitative Feedback:** Gather feedback from potential users through surveys or small focus groups on ease of use, perceived value, and areas for improvement.
- Improve the literature review and comparative summary table and use proper IEEE reference style.

22FYDP17 – QUEST Alumni Portal

Roll Nos.: 22SW54, 22SW26
Supervisor: Dr. Imtiaz Halepoto
Comments / Suggestions:

- Limit project scope due to the large administrative and legal domain.
- Specify tools, methods, and technologies clearly.
- Provide a detailed implementation roadmap.
- Improve the literature review and comparative summary table and use proper IEEE reference style.
- Include valuable services for alumni:

- Exclusive Services: Offer perks like transcript requests, degree verification services, or library access directly through the portal.
- Job Board for "Freshers": Focus the job board specifically on entry-level roles where alumni specifically want to hire juniors from their own university (high trust).
- Events: Integrate an event management feature for on-campus alumni reunions or guest lectures.

Dr. Ali Raza Bhangwar

FYDP Coordinator